LIBRARY MANAGEMENT SYSTEM

PROGRAM

```
SET SERVEROUTPUT ON;
SET LINESIZE 150;
SET PAGESIZE 100;

CREATE TABLE categories (
   category_id NUMBER PRIMARY KEY,
   category_name VARCHAR2(50) NOT NULL
);
```

OUTPUT

```
Table CATEGORIES created.
```

```
CREATE TABLE authors (
   author_id NUMBER PRIMARY KEY,
   author_name VARCHAR2(100) NOT NULL
);
```

OUTPUT

```
Table AUTHORS created.
```

```
CREATE TABLE books (
   book_id NUMBER PRIMARY KEY,
   title VARCHAR2(150) NOT NULL,
   author_id NUMBER REFERENCES authors(author_id),
   category_id NUMBER REFERENCES categories(category_id),
   publication_year NUMBER(4),
   total_copies NUMBER DEFAULT 1,
   available_copies NUMBER DEFAULT 1
);
```

OUTPUT

```
Table BOOKS created.
```

```
CREATE TABLE members (
   member_id NUMBER PRIMARY KEY,
   member_name VARCHAR2(100) NOT NULL,
   email VARCHAR2(100) UNIQUE,
   phone VARCHAR2(15),
   join_date DATE DEFAULT SYSDATE
);
```

OUTPUT

Table MEMBERS created.

```
CREATE TABLE loans (
   loan_id NUMBER PRIMARY KEY,
   book_id NUMBER REFERENCES books(book_id),
   member_id NUMBER REFERENCES members(member_id),
   issue_date DATE DEFAULT SYSDATE,
   due_date DATE,
   return_date DATE
);
```

OUTPUT
Table LOANS created.

```
CREATE SEQUENCE category_seq
START WITH 1
INCREMENT BY 1;

CREATE SEQUENCE author_seq
START WITH 1
INCREMENT BY 1;

CREATE SEQUENCE book_seq
START WITH 1
INCREMENT BY 1;

CREATE SEQUENCE member_seq
START WITH 1
INCREMENT BY 1;

CREATE SEQUENCE loan_seq
START WITH 1
INCREMENT BY 1;
```

OUTPUT
Sequence CATEGORY_SEQ created.
Sequence AUTHOR_SEQ created.
Sequence BOOK_SEQ created.
Sequence MEMBER_SEQ created.
Sequence LOAN_SEQ created.

```
INSERT INTO categories
VALUES (category_seq.NEXTVAL, 'Programming');
```

```
INSERT INTO categories
VALUES (category_seq.NEXTVAL, 'Database');

INSERT INTO categories
VALUES (category_seq.NEXTVAL, 'Fiction');

INSERT INTO categories
VALUES (category_seq.NEXTVAL, 'Science');

INSERT INTO categories
VALUES (category_seq.NEXTVAL, 'History');
```

```
OUTPUT
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
```

```
INSERT INTO authors
VALUES (author_seq.NEXTVAL, 'Robert C. Martin');

INSERT INTO authors
VALUES (author_seq.NEXTVAL, 'Ramez Elmasri');

INSERT INTO authors
VALUES (author_seq.NEXTVAL, 'George Orwell');

INSERT INTO authors
VALUES (author_seq.NEXTVAL, 'Stephen Hawking');

INSERT INTO authors
VALUES (author_seq.NEXTVAL, 'Yuval Noah Harari');
```

```
OUTPUT
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
```

```
INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
```

```
    'Clean Code',
    1,
    1,
    2008,
    5,
    4
);

INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
    'Database Systems',
    2,
    2,
    2016,
    4,
    3
);

INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
    '1984',
    3,
    3,
    1949,
    6,
    5
);

INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
    'A Brief History of Time',
    4,
    4,
    1988,
    3,
    2
);

INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
```

```
    'Sapiens',
    5,
    5,
    2011,
    5,
    5
);

INSERT INTO books
VALUES (
    book_seq.NEXTVAL,
    'The Art of Computer Programming',
    1,
    1,
    2000,
    2,
    2
);

OUTPUT
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.

INSERT INTO members
VALUES (
    member_seq.NEXTVAL,
    'Arun Kumar',
    'arun@gmail.com',
    '9876543210',
    DATE '2025-01-10'
);

INSERT INTO members
VALUES (
    member_seq.NEXTVAL,
    'Priya Sharma',
    'priya@gmail.com',
    '9876543211',
    DATE '2025-02-15'
);
```

```
INSERT INTO members
VALUES (
   member_seq.NEXTVAL,
   'Rahul Das',
   'rahul@gmail.com',
   '9876543212',
   DATE '2025-03-20'
);

INSERT INTO members
VALUES (
   member_seq.NEXTVAL,
   'Sneha Rao',
   'sneha@gmail.com',
   '9876543213',
   DATE '2025-04-05'
);
```

OUTPUT

```
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
```

```
INSERT INTO loans
VALUES (
   loan_seq.NEXTVAL,
   1,
   1,
   DATE '2025-06-01',
   DATE '2025-06-15',
   DATE '2025-06-12'
);

INSERT INTO loans
VALUES (
   loan_seq.NEXTVAL,
   2,
   2,
   DATE '2025-06-05',
   DATE '2025-06-19',
   NULL
);
```

```
INSERT INTO loans
VALUES (
   loan_seq.NEXTVAL,
   3,
   3,
   DATE '2025-06-10',
   DATE '2025-06-24',
   DATE '2025-06-20'
);

INSERT INTO loans
VALUES (
   loan_seq.NEXTVAL,
   4,
   4,
   DATE '2025-06-12',
   DATE '2025-06-26',
   NULL
);

COMMIT;
```

```
OUTPUT
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Commit complete.
```

```
SELECT
   book_id,
   title,
   publication_year,
   total_copies,
   available_copies
FROM books
ORDER BY book_id;
```

```
OUTPUT
--
  BOOK_ID TITLE                          PUBLICATION_YEAR TOTAL_COPIES AVAILABLE_COPIES
----------------------------------------------------------------------------------
        1 Clean Code                     2008                5             4
```

2 Database Systems 2016 4 3
3 1984 1949 6 5
4 A Brief History of Time 1988 3 2
5 Sapiens 2011 5 5
6 The Art of Computer Programming 2000 2 2

```
SELECT
   b.book_id,
   b.title,
   a.author_name,
   c.category_name,
   b.publication_year,
   b.available_copies
FROM books b
JOIN authors a
ON b.author_id = a.author_id
JOIN categories c
ON b.category_id = c.category_id
ORDER BY b.book_id;
```

OUTPUT

```
--
1       Clean Code
Robert C. Martin     Programming
2008            4
--
2       Database Systems
Ramez Elmasri        Database
2016            3
--
3       1984
George Orwell        Fiction
1949            5
--
4       A Brief History of Time
Stephen Hawking      Science
1988            2
--
5       Sapiens
Yuval Noah Harari    History
2011            5
--
6       The Art of Computer Programming
Robert C. Martin     Programming
```

2000 2

```
SELECT
   member_id,
   member_name,
   email,
   phone,
   join_date
FROM members
ORDER BY member_id;
```

OUTPUT

```
--
MEMBER_ID MEMBER_NAME    EMAIL            PHONE       JOIN_DATE
-----------------------------------------
1         Arun Kumar     arun@gmail.com    9876543210  10-JAN-25
2         Priya Sharma   priya@gmail.com   9876543211  15-FEB-25
3         Rahul Das      rahul@gmail.com   9876543212  20-MAR-25
4         Sneha Rao      sneha@gmail.com   9876543213  05-APR-25
```

```
SELECT
   book_id,
   title,
   available_copies
FROM books
WHERE available_copies > 0
ORDER BY book_id;
```

OUTPUT

```
--
BOOK_ID  TITLE                          AVAILABLE_COPIES
-----------------------------------------------
1        Clean Code                     4
2        Database Systems               3
3        1984                           5
4        A Brief History of Time        2
5        Sapiens                        5
6        The Art of Computer Programming  2
```

```
SELECT
   book_id,
   title
FROM books
WHERE LOWER(title) LIKE '%computer%';
```

OUTPUT

```
--
BOOK_ID  TITLE
-----------------------------------
6        The Art of Computer Programming
```

```
SELECT
   title,
   publication_year
FROM books
WHERE publication_year > 2000
ORDER BY publication_year;
```

OUTPUT

```
--
TITLE                PUBLICATION_YEAR
-----------------------------------
Clean Code              2008
Sapiens                 2011
Database Systems        2016
```

```
SELECT COUNT(*) AS total_book_titles
FROM books;
```

OUTPUT

```
--
TOTAL_BOOK_TITLES
-----------------
6
```

```
SELECT SUM(total_copies) AS total_copies
FROM books;
```

OUTPUT

```
--
TOTAL_COPIES
------------
25
```

```
SELECT SUM(available_copies) AS available_copies
FROM books;
```

OUTPUT

```
--
AVAILABLE_COPIES
----------------
21

SELECT COUNT(*) AS total_members
FROM members;

OUTPUT
--
TOTAL_MEMBERS
-------------
4

SELECT
   c.category_name,
   COUNT(b.book_id) AS number_of_books
FROM categories c
LEFT JOIN books b
ON c.category_id = b.category_id
GROUP BY c.category_name
ORDER BY c.category_name;

OUTPUT
--
CATEGORY_NAME    NUMBER_OF_BOOKS
--------------------------
Database         1
Fiction          1
History          1
Programming      2
Science          1

SELECT
   l.loan_id,
   m.member_name,
   b.title,
   l.issue_date,
   l.due_date
FROM loans l
JOIN members m
ON l.member_id = m.member_id
JOIN books b
ON l.book_id = b.book_id
```

```
WHERE l.return_date IS NULL
ORDER BY l.loan_id;

OUTPUT
--
LOAN_ID  MEMBER_NAME    TITLE                ISSUE_DATE  DUE_DATE
-----------------------------------------------
2        Priya Sharma   Database Systems       05-JUN-25   19-JUN-25
4        Sneha Rao      A Brief History of Time  12-JUN-25   26-JUN-25

SELECT
   l.loan_id,
   m.member_name,
   b.title,
   l.issue_date,
   l.due_date,
   l.return_date
FROM loans l
JOIN members m
ON l.member_id = m.member_id
JOIN books b
ON l.book_id = b.book_id
ORDER BY l.loan_id;

OUTPUT
--
LOAN_ID  MEMBER_NAME    TITLE                ISSUE_DATE  DUE_DATE    RETURN_DATE
-------------------------------------------------------
1        Arun Kumar     Clean Code             01-JUN-25   15-JUN-25   12-JUN-25
2        Priya Sharma   Database Systems       05-JUN-25   19-JUN-25
3        Rahul Das      1984                   10-JUN-25   24-JUN-25   20-JUN-25
4        Sneha Rao      A Brief History of Time  12-JUN-25   26-JUN-25

SELECT
   m.member_name,
   COUNT(l.loan_id) AS books_borrowed
FROM members m
LEFT JOIN loans l
ON m.member_id = l.member_id
GROUP BY m.member_id, m.member_name
ORDER BY m.member_id;

OUTPUT
--
```

```
MEMBER_NAME    BOOKS_BORROWED
----------------------
Arun Kumar     1
Priya Sharma   1
Rahul Das      1
Sneha Rao      1
```

```
SELECT
   b.book_id,
   b.title
FROM books b
LEFT JOIN loans l
ON b.book_id = l.book_id
WHERE l.loan_id IS NULL
ORDER BY b.book_id;
```

```
OUTPUT
--
BOOK_ID  TITLE
-------------------------------------
5        Sapiens
6        The Art of Computer Programming
```

```
SELECT
   l.loan_id,
   m.member_name,
   b.title,
   l.due_date
FROM loans l
JOIN members m
ON l.member_id = m.member_id
JOIN books b
ON l.book_id = b.book_id
WHERE l.return_date IS NULL
AND l.due_date < TRUNC(SYSDATE)
ORDER BY l.loan_id;
```

```
OUTPUT FROM ORIGINAL RUN
--
LOAN_ID  MEMBER_NAME    TITLE                DUE_DATE
----------------------------------------
2        Priya Sharma   Database Systems         19-JUN-25
4        Sneha Rao      A Brief History of Time  26-JUN-25
```

```
SELECT
   l.loan_id,
   m.member_name,
   b.title,
   l.due_date,
   CASE
      WHEN NVL(l.return_date, TRUNC(SYSDATE)) > l.due_date
      THEN
         (NVL(l.return_date, TRUNC(SYSDATE))
          - l.due_date) * 5
      ELSE 0
   END AS fine_amount
FROM loans l
JOIN members m
ON l.member_id = m.member_id
JOIN books b
ON l.book_id = b.book_id
ORDER BY l.loan_id;

OUTPUT FROM ORIGINAL RUN
--
LOAN_ID  MEMBER_NAME    TITLE                  FINE_AMOUNT
-----------------------------------------
1        Arun Kumar     Clean Code                    0
2        Priya Sharma   Database Systems              2315
3        Rahul Das      1984                          0
4        Sneha Rao      A Brief History of Time       2280
--
NOTE:
The fine for active loans depends on SYSDATE.
These values change when the script is executed on a different date.

DECLARE
   v_books NUMBER;
   v_copies NUMBER;
   v_available NUMBER;
   v_members NUMBER;
   v_loans NUMBER;
BEGIN

   SELECT COUNT(*)
   INTO v_books
   FROM books;
```

```
    SELECT SUM(total_copies)
    INTO v_copies
    FROM books;

    SELECT SUM(available_copies)
    INTO v_available
    FROM books;

    SELECT COUNT(*)
    INTO v_members
    FROM members;

    SELECT COUNT(*)
    INTO v_loans
    FROM loans
    WHERE return_date IS NULL;

    DBMS_OUTPUT.PUT_LINE('======================================');
    DBMS_OUTPUT.PUT_LINE('      LIBRARY MANAGEMENT SYSTEM');
    DBMS_OUTPUT.PUT_LINE('======================================');
    DBMS_OUTPUT.PUT_LINE('Total Book Titles : ' || v_books);
    DBMS_OUTPUT.PUT_LINE('Total Copies      : ' || v_copies);
    DBMS_OUTPUT.PUT_LINE('Available Copies  : ' || v_available);
    DBMS_OUTPUT.PUT_LINE('Total Members     : ' || v_members);
    DBMS_OUTPUT.PUT_LINE('Active Loans      : ' || v_loans);
    DBMS_OUTPUT.PUT_LINE('======================================');

END;
/

OUTPUT
--
======================================
      LIBRARY MANAGEMENT SYSTEM
======================================
Total Book Titles : 6
Total Copies      : 25
Available Copies  : 21
Total Members     : 4
Active Loans      : 2
======================================

DECLARE
```

```
    CURSOR book_cursor IS
        SELECT
            b.title,
            a.author_name
        FROM books b
        JOIN authors a
        ON b.author_id = a.author_id
        ORDER BY b.book_id;

BEGIN

    DBMS_OUTPUT.PUT_LINE('');
    DBMS_OUTPUT.PUT_LINE('========== BOOK LIST ==========');

    FOR r IN book_cursor
    LOOP
        DBMS_OUTPUT.PUT_LINE(
            r.title || ' - ' || r.author_name
        );
    END LOOP;

END;
/

OUTPUT
--
========== BOOK LIST ==========
Clean Code - Robert C. Martin
Database Systems - Ramez Elmasri
1984 - George Orwell
A Brief History of Time - Stephen Hawking
Sapiens - Yuval Noah Harari
The Art of Computer Programming - Robert C. Martin

CREATE OR REPLACE VIEW book_details AS
SELECT
    b.book_id,
    b.title,
    a.author_name,
    c.category_name,
    b.publication_year,
    b.total_copies,
    b.available_copies
FROM books b
```

```
JOIN authors a
ON b.author_id = a.author_id
JOIN categories c
ON b.category_id = c.category_id;
```

OUTPUT
View BOOK_DETAILS created.

```
SELECT *
FROM book_details
ORDER BY book_id;
```

OUTPUT

```
--
BOOK_ID  TITLE                          AUTHOR_NAME
--------------------------------------------------
1        Clean Code                     Robert C. Martin
2        Database Systems               Ramez Elmasri
3        1984                           George Orwell
4        A Brief History of Time        Stephen Hawking
5        Sapiens                        Yuval Noah Harari
6        The Art of Computer Programming   Robert C. Martin
```

```
CREATE OR REPLACE VIEW active_loans AS
SELECT
   l.loan_id,
   m.member_name,
   b.title,
   l.issue_date,
   l.due_date
FROM loans l
JOIN members m
ON l.member_id = m.member_id
JOIN books b
ON l.book_id = b.book_id
WHERE l.return_date IS NULL;
```

OUTPUT
View ACTIVE_LOANS created.

```
SELECT *
FROM active_loans
ORDER BY loan_id;
```

OUTPUT

```
--
LOAN_ID  MEMBER_NAME    TITLE                 ISSUE_DATE  DUE_DATE
-----------------------------------------------
2        Priya Sharma   Database Systems        05-JUN-25   19-JUN-25
4        Sneha Rao      A Brief History of Time  12-JUN-25   26-JUN-25
```

```
CREATE OR REPLACE FUNCTION calculate_fine(
   p_loan_id NUMBER
)
RETURN NUMBER
IS
   v_due_date DATE;
   v_return_date DATE;
   v_fine NUMBER;
BEGIN

   SELECT
      due_date,
      return_date
   INTO
      v_due_date,
      v_return_date
   FROM loans
   WHERE loan_id = p_loan_id;

   IF NVL(v_return_date, TRUNC(SYSDATE)) > v_due_date THEN

      v_fine :=
         (NVL(v_return_date, TRUNC(SYSDATE))
         - v_due_date) * 5;

   ELSE

      v_fine := 0;

   END IF;

   RETURN v_fine;

EXCEPTION
   WHEN NO_DATA_FOUND THEN
      RETURN 0;
END;
```

/

OUTPUT
Function CALCULATE_FINE compiled.

```
SELECT
   loan_id,
   calculate_fine(loan_id) AS fine
FROM loans
ORDER BY loan_id;
```

OUTPUT FROM ORIGINAL RUN

```
--
LOAN_ID  FINE
-------
1       0
2       2315
3       0
4       2280
```

```
CREATE OR REPLACE PROCEDURE issue_book(
   p_book_id NUMBER,
   p_member_id NUMBER
)
IS
   v_available NUMBER;
BEGIN

   SELECT available_copies
   INTO v_available
   FROM books
   WHERE book_id = p_book_id;

   IF v_available > 0 THEN

      INSERT INTO loans
      (
         loan_id,
         book_id,
         member_id,
         issue_date,
         due_date
      )
      VALUES
```

```
        (
            loan_seq.NEXTVAL,
            p_book_id,
            p_member_id,
            TRUNC(SYSDATE),
            TRUNC(SYSDATE) + 14
        );

        UPDATE books
        SET available_copies = available_copies - 1
        WHERE book_id = p_book_id;

        COMMIT;

        DBMS_OUTPUT.PUT_LINE(
            'Book issued successfully.'
        );

    ELSE

        DBMS_OUTPUT.PUT_LINE(
            'Book is not available.'
        );

    END IF;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE(
            'Book does not exist.'
        );
END;
/
```

OUTPUT

```
Procedure ISSUE_BOOK compiled.
```

```
BEGIN
    issue_book(5, 1);
END;
/
```

OUTPUT

```
Book issued successfully.
```

```
CREATE OR REPLACE PROCEDURE return_book(
   p_loan_id NUMBER
)
IS
   v_book_id NUMBER;
   v_return_date DATE;
BEGIN

   SELECT
      book_id,
      return_date
   INTO
      v_book_id,
      v_return_date
   FROM loans
   WHERE loan_id = p_loan_id;

   IF v_return_date IS NULL THEN

      UPDATE loans
      SET return_date = TRUNC(SYSDATE)
      WHERE loan_id = p_loan_id;

      UPDATE books
      SET available_copies = available_copies + 1
      WHERE book_id = v_book_id;

      COMMIT;

      DBMS_OUTPUT.PUT_LINE(
         'Book returned successfully.'
      );

   ELSE

      DBMS_OUTPUT.PUT_LINE(
         'Book was already returned.'
      );

   END IF;

EXCEPTION
   WHEN NO_DATA_FOUND THEN
```

```
        DBMS_OUTPUT.PUT_LINE(
            'Loan record does not exist.'
        );
END;
/
```

OUTPUT

Procedure RETURN_BOOK compiled.

```
BEGIN
    return_book(2);
END;
/
```

OUTPUT

Book returned successfully.

```
SELECT
    b.book_id,
    b.title,
    a.author_name,
    c.category_name,
    b.total_copies,
    b.available_copies
FROM books b
JOIN authors a
ON b.author_id = a.author_id
JOIN categories c
ON b.category_id = c.category_id
ORDER BY b.book_id;
```

OUTPUT

```
--
BOOK_ID  TITLE                          AUTHOR_NAME
--------------------------------------------------
CATEGORY_NAME    TOTAL_COPIES  AVAILABLE_COPIES
-----------------------------------
--
1        Clean Code                     Robert C. Martin
Programming       5            4
--
2        Database Systems               Ramez Elmasri
Database          4            4
--
```

```
3       1984                        George Orwell
Fiction         6           5
--
4       A Brief History of Time         Stephen Hawking
Science         3           2
--
5       Sapiens                     Yuval Noah Harari
History         5           4
--
6       The Art of Computer Programming   Robert C. Martin
Programming       2           2

BEGIN

   DBMS_OUTPUT.PUT_LINE('');
   DBMS_OUTPUT.PUT_LINE('====================================');
   DBMS_OUTPUT.PUT_LINE(' LIBRARY PROJECT EXECUTED SUCCESSFULLY');
   DBMS_OUTPUT.PUT_LINE('====================================');

END;
/

OUTPUT
--
====================================
 LIBRARY PROJECT EXECUTED SUCCESSFULLY
====================================

COMMIT;

OUTPUT
Commit complete.
```